

ICCT MIS Capstone - Revised

Angelica Padil, Bryan Soliven, Richard Erenea, Ronjay Pulido, John Harold Uvero, Mikaela Montealto, Melody Rances, Arlyn Cenidoza, Grace Joy Flores

May 2026

Abstract

The borrowing and returning of resources at ICCT Cainta Campus currently relies on manual request letters or faculty ID presentation, which is inefficient, time-consuming, and lacks transparency in tracking availability. Professors often experience delays in borrowing, while MIS faculty struggle with monitoring overdue returns and managing resource quantities. This study developed a QR code-enabled web-based automated resource borrowing and returning system to replace the manual request process and streamline transactions. Through QR codes placed on physical resources or resource room entrances, professors can scan and be directed to a web-based dashboard where they can view resource availability and submit borrowing requests. QR codes serve as a convenient shortcut to access the system, while actual user authentication is handled through a secure email-based login system. Borrowing requests are submitted digitally and require administrator approval, eliminating the need for physical request letters or ID presentation. The system automatically tracks borrowed items, highlights due and overdue returns on the dashboard, and sends real-time notifications to administrators and faculty. Key features include a room or delivery location field for item delivery routing, automated notification system with real-time polling, and borrow history tracking. By integrating QR code technology and automation, the developed solution reduces manual workload, accelerates transactions, and enhances transparency in resource management. The expected contribution of this study is an IT-driven solution that modernizes the borrowing and returning process, thereby improving academic support services for MIS faculty at ICCT Cainta Campus.

Keywords — QR Code Technology, Web-Based System, Resource Management, Automated Borrowing and Returning, Laravel, Information System

I. Introduction

A. Background and Motivation

Resource borrowing and returning is an essential support function in academic institutions, particularly for faculty members who rely on shared materials and equipment to deliver instruction effectively. At ICCT Cainta Campus, professors in the MIS department currently request resources either by submitting a formal request letter or by presenting their faculty ID. While this process ensures accountability, it is slow, manual, and prone to delays.

MIS faculty also face challenges in monitoring overdue returns and tracking the availability of resources, as there is no centralized system that provides real-time visibility of inventory. These inefficiencies disrupt schedules, reduce productivity, and complicate resource management.

In the field of Information Technology, automation and QR code technology have been widely adopted to streamline workflows, improve accuracy, and enhance transparency. This motivates the development of a QR code-enabled web-based system that modernizes the borrowing and returning process for MIS faculty at ICCT Cainta Campus.

B. Problem Statement

The existing borrowing and returning process at ICCT Cainta Campus is inefficient and outdated. Professors must either submit a request letter or present their faculty ID to borrow resources, which consumes time and lacks transparency. MIS faculty struggle to monitor overdue returns and resource availability, as there is no centralized dashboard to track inventory in real time. There is also no automated notification system to alert faculty about due dates or overdue items, and no mechanism to inform staff where borrowed items should be delivered. These limitations result in delays, poor accountability, and reduced efficiency in resource management.

C. Objectives of the Study

The general objective of this study is to design and develop a QR code-enabled web-based system that automates the borrowing and returning of resources, replacing the current manual process of request letters and faculty ID presentation.

Specifically, the study achieved the following objectives:

1. Implement QR code integration that directs professors to the web-based system for resource transactions.
2. Display available resources and their exact quantities in real time on a centralized dashboard.
3. Allow professors to submit borrowing requests digitally with a delivery room or location field, eliminating the need for request letters or physical ID presentation.
4. Enable administrators to approve, reject, and monitor borrowing requests, including tracking due and overdue returns through the dashboard.
5. Provide a notification system that alerts both faculty and administrators about approval status, due dates, and overdue items directly within the system dashboard.
6. Maintain a searchable borrow history for both faculty and administrators to track past transactions.

D. Scope of the Study

The scope defines the specific areas, users, and functionalities that the developed system covers:

- **Target Users:** The system is specifically designed for the MIS faculty and administrators at the ICCT Cainta Campus.
- **QR Code Integration:** QR codes placed on resources or at resource room entrances redirect professors to the web-based

system. The QR code serves as a convenient access shortcut.

- **Real-Time Resource Management:** The system provides real-time tracking and display of available resources and their exact quantities.
- **Digital Workflow Submission:** Professors can submit borrowing requests digitally, effectively replacing the need for manual request letters. Each request includes a room or delivery location field to inform staff where to deliver the item.
- **Administrative Oversight:** Administrators are equipped with tools to approve or reject borrowing requests, view active borrows, and monitor due and overdue returns through a dedicated dashboard.
- **Automated Notifications:** The system generates automated alerts displayed within the dashboard to notify both users and administrators regarding approval status, upcoming due dates, and overdue items. A bell icon with an unread count badge provides real-time visibility of pending notifications.
- **Borrow History:** Both faculty and administrators can search and view past borrowing transactions.
- **Technical Platform:** The solution is a web-based system built with Laravel framework and MySQL database, accessible via standard web browsers. Role-based access control differentiates faculty and administrator privileges. Authentication is handled through secure email-based login.

E. Limitations of the Study

The limitations identify the boundaries and external factors that the system cannot control or will not address:

- **Connectivity Requirements:** The system is entirely dependent on internet connectivity to function and synchronize data in real time.
- **Account Credential Dependency:** The system relies on registered user accounts created by administrators. Password recovery requires administrator intervention.
- **Hardware Dependency:** Users must have access to web-enabled devices (such as smartphones, tablets, or computers) and optionally QR code scanning devices to interact with the system.
- **Maintenance Boundaries:** The scope excludes the physical maintenance of resources; the system only tracks their digital status and availability.
- **Infrastructure Constraints:** The system may face challenges in environments with bandwidth limitations or specific device compatibility issues.
- **Geographic and Departmental Focus:** The system is currently deployed for the MIS faculty at the ICCT Cainta Campus, without addressing scalability for other departments or locations.
- **Data Integrity:** The system relies on the honesty of the borrower to report the actual condition of the resource upon return, as there is no automated hardware sensor to detect physical damage.
- **Delivery Location Tracking:** The system requires faculty to manually input their room or delivery location, which may result in incomplete or inaccurate entries if not properly filled out.
- **In-App Notifications Only:** All alerts, including approval status, due dates, overdue reminders, and return confirmations, are displayed only within the system dashboard. Users must log in to the platform to view these notifications.

II. Related Works

The continuous advancement of information technology has reshaped how academic institutions manage student identification, attendance monitoring, and resource utilization. Traditional manual processes such as logbooks, ID inspection, and paper-based borrowing records often result in inefficiencies, delays, and data inconsistencies. Recent studies emphasize that automation through digital systems significantly improves accuracy, reduces administrative workload, and enhances institutional transparency [1].

QR code technology has become one of the most practical tools for implementing automated monitoring systems in educational environments. According to Roy [1], QR-based attendance systems offer a low-cost yet reliable alternative to manual attendance tracking. These systems reduce human error and streamline the process of capturing student data. Similarly, Masalha and Hirzallah [2] demonstrated that QR code scanning enables faster attendance recording while maintaining data accuracy and accessibility for administrators.

In addition to attendance monitoring, QR codes have been applied in broader academic resource management contexts. Rahman et al. [3] explain that QR technology can be effectively used for tracking borrowed materials such as laboratory tools and library books. Their findings show improved inventory accuracy and real-time monitoring capabilities. By digitizing borrowing records, institutions gain structured data that can support planning and decision-making processes.

The effectiveness of QR systems also depends on system design and quality standards. Liew and Tan [4] highlighted the importance of authentication accuracy and fraud prevention in QR-based attendance applications. Meanwhile, Sharma et al. [5] stressed that academic information systems must comply with quality models such as ISO/IEC 25010 to ensure usability, reliability, performance efficiency, and compatibility across devices. Without proper system design considerations, digital platforms may face adoption challenges among users.

Several foreign studies have explored integrated systems that combine QR-based login authentication with resource management functions. Singh et al. [11] developed a QR-based access control and resource tracking system that reduced manual processing time and minimized record errors. Hariyanto et al. [14] implemented a QR student management system featuring login verification and asset checkout workflows. Mahmood et al. [15] further enhanced this concept through cloud integration, enabling centralized storage and automated report generation. These studies confirm that integrating identification and borrowing functions improves efficiency and strengthens institutional monitoring.

Within the Philippine context, QR-based systems have been widely adopted for attendance monitoring and ID validation. Local studies report improvements in efficiency, user acceptability, and administrative workload reduction [6], [7]. Research on QR and SMS-based attendance monitoring also highlights enhanced punctuality and accountability among students [8], [10]. Additionally, digital ID validation systems have demonstrated faster verification procedures and improved security compared to traditional manual inspection methods [9]. A QR-integrated library circulation system implemented in a local campus further showed improved transaction speed and record accuracy [16].

Despite the proven effectiveness of QR-based systems, current implementations reveal several limitations. Many focus on isolated functions such as attendance tracking or library circulation, without providing a comprehensive solution that integrates authentication, resource availability tracking, borrowing management, delivery tracking, and real-time notifications on one platform. Foreign-developed systems often assume robust technological infrastructure, which may not align with local institutional constraints such as bandwidth and device compatibility. Local studies, while effective in specific contexts, lack scalability across multiple departments and do not fully address overdue monitoring, delivery routing, or centralized reporting.

Research Gap

These gaps highlight the need for a centralized QR code-based faculty authentication and resource borrowing management system tailored to the operational environment of ICCT Cainta Campus. By integrating QR code access, availability tracking, digital borrowing requests with delivery location input, administrator approval workflows, and real-time notifications into a single web-based platform, the developed system aims to enhance efficiency, accountability, and institutional digital transformation for MIS faculty.

III. Methodology

A. Research Design

This study adopts a developmental research design, which focuses on the systematic analysis, design, development, and implementation of an information system to address identified inefficiencies. Developmental research is appropriate because the study aims to create a new IT solution based on existing problems in the borrowing and returning process at ICCT Cainta Campus. The methodology emphasizes the full software development lifecycle, ensuring that the developed solution is grounded in user needs and technical feasibility.

The Iterative Software Development Life Cycle (SDLC) model was used as the development framework, allowing for continuous refinement of system features based on testing and feedback throughout the development process.

Figure 1: Iterative SDLC Model Used for System Development

Insert Fig. 1: Iterative SDLC Model here

Iterative SDLC Model

Fig. 1 shows the Iterative SDLC phases followed in developing the QR Resource System. The cycle includes Planning, Analysis, Design, Implementation, Testing, and Deployment, with each iteration allowing for refinement based on feedback.

B. Conceptual Framework

The proposed system follows an enhanced Input-Process-Output (IPO) model with a feedback loop to support continuous improvement and monitoring. This framework illustrates how user actions are transformed into structured outputs through system processes, ensuring efficiency, accountability, and transparency in resource management.

Input includes faculty credentials, web-enabled devices, the faculty database, the resource inventory database, and the administrator control panel, which serve as the primary data sources and system components.

Process represents the system operations, including user authentication, resource availability checking, digital borrowing request submission with room/location input, administrator approval and rejection workflows, transaction recording with automatic timestamping, real-time inventory status updates, due-date monitoring, automated notification generation, and borrow history tracking.

Output presents the system results, such as authenticated faculty access, approved borrowing and return transactions, updated resource inventory status, delivery location information, overdue alerts and notifications, and searchable borrow records.

Feedback provides borrowing trends, late return analysis, resource utilization insights, and system performance evaluation, which are used to improve system efficiency, accuracy, and overall performance.

Figure 2: Conceptual Framework Using the IPO Model

Insert Fig. 2: IPO Conceptual Framework here

IPO Model Conceptual Framework

Fig. 2 illustrates the Input-Process-Output (IPO) model with a feedback mechanism for the developed QR Code-Enabled Faculty Resource Borrowing and Returning System.

C. System Flowchart

The system flowchart illustrates how the two types of users—Admin and Faculty—interact with the system.

Figure 3: System Flowchart for Admin and Faculty Users

Insert Fig. 3: System Flowchart here

System Flowchart

Fig. 3 shows the workflow of the system from both Admin and Faculty perspectives. The Admin path includes login, dashboard access, approval/rejection of requests, and monitoring of active and overdue borrows. The Faculty path includes login, resource browsing, borrow request submission (with room/location input), and borrow history viewing.

D. Proposed System Overview

The developed solution is a QR code-enabled web-based borrowing and returning system designed for MIS faculty at ICCT Cainta Campus. Professors can scan a QR code placed on physical resources or at the resource room to access the system, where they can view

available resources and their quantities in real time. Borrowing requests are submitted digitally through the system with a room or delivery location field, replacing the current manual process of request letters or ID presentation. Administrators review and approve or reject requests, monitor active, due, and overdue returns, and manage resource inventory.

The system's major components include:

- QR code access integration
- Resource availability dashboard
- Digital borrowing request workflow with delivery location input
- Administrator approval and rejection system
- Real-time notification system
- Borrow history and status tracking
- Role-based authentication for faculty and administrators

Intended users are MIS faculty, administrators, and professors at ICCT Cainta Campus.

E. Requirements Analysis

The requirements of the developed QR code-enabled web-based borrowing and returning system were identified based on the operational needs of MIS faculty and administrators at ICCT Cainta Campus, as well as technical constraints of the institutional environment.

Functional Requirements

1. QR code scanning and redirecting to the system dashboard.
2. Real-time display of available resources and quantities.
3. Digital borrowing request submission with room or delivery location input.
4. Administrator approval and rejection of borrowing requests.
5. Real-time notifications for approval status, due dates, and overdue returns.
6. Borrow history tracking with search functionality.
7. Secure login via email and password.

Non-Functional Requirements

1. **Authentication Security:** Role-based login system ensuring only verified faculty and administrators can access their respective dashboards.
2. **Web-based Accessibility:** Accessible via standard web browsers on desktop and mobile devices.
3. **Role-Based Access Control:** Separate interfaces and permissions for faculty and administrator roles.
4. **Real-Time Data Synchronization:** Live updates of resource availability and notification counts.
5. **User-Friendly Interface:** Clean, intuitive design to encourage adoption.
6. **Compatibility:** Compatible with existing institutional infrastructure.

F. Ethical, Legal, and Technical Considerations

From an ethical perspective, the developed system ensures fairness, transparency, and accountability in the borrowing and returning of institutional resources. By recording all transactions with timestamps and maintaining accurate records, the system prevents disputes and misuse while promoting equal access among faculty members. Ethical responsibility also extends to designing a user-friendly interface that minimizes errors and supports adoption.

Legally, the system complies with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173) and institutional policies governing information management. Faculty data, borrowing records, and transaction logs are handled with strict confidentiality and protected against unauthorized access through role-based permissions. Compliance also includes adherence to institutional IT governance standards, including rules on access control, record retention, and proper authorization of borrowing requests.

On the technical side, the system implements secure authentication mechanisms through email-based login with hashed passwords. A role-based access control (RBAC) system differentiates faculty and administrator privileges. Data storage uses secure database practices with parameterized queries to prevent SQL injection. The system is built on the Laravel framework, which includes built-in protection against common web vulnerabilities such as cross-site scripting (XSS), cross-site request forgery (CSRF), and SQL injection.

IV. System Development and Features

A. Technology Stack

The system was developed using the following technologies:

- **Backend Framework:** Laravel 11 (PHP 8.1+)
- **Database:** MySQL
- **Frontend:** HTML5, CSS3 (Tailwind CSS), JavaScript
- **Web Server:** Apache
- **Version Control:** Git
- **Development Platform:** Ubuntu Linux

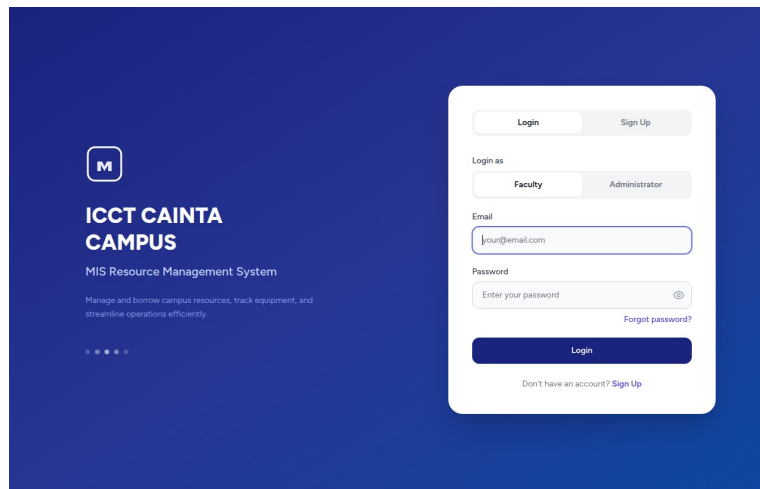
B. System Modules

1. Authentication Module

The system implements a role-based authentication system. Users log in using their registered email and password. Two roles are supported:

- **Administrator:** Has full access to manage resources, approve/reject borrow requests, view all transactions, and manage users.
- **Faculty:** Can view available resources, submit borrow requests, and view their own borrow history.

Figure 4: Login Page



Login Page

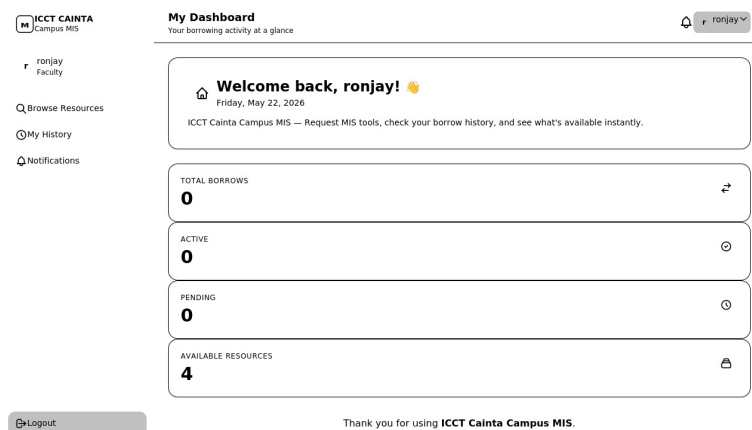
Fig. 4 shows the login page where users authenticate using their registered email and password.

2. Faculty Dashboard

Upon logging in, faculty members are directed to their dashboard, which displays:

- Total number of available resources
- Pending borrow requests
- Active (approved) borrows
- Due and overdue items with visual indicators
- Recently borrowed items

Figure 5: Faculty Dashboard



Faculty Dashboard

Fig. 5 shows the faculty dashboard with overview statistics and resource availability.

3. Resource Management

Faculty can browse available resources with real-time quantity display. Each resource shows its name, category, description, and available units. Resources with zero available units are clearly marked as unavailable.

Figure 6: Browse Resources Page

ICCT CAINTA
Campus MES

ronjay
Faculty

Browse Resources

My History

Notifications

Logout

Browse Resources
View and borrow available campus resources

AVAILABLE
40

IN USE
0

UNAVAILABLE
0

Available

HDMI Cable
Standard HDMI Cable for classroom use

10/10

+ Borrow This Item

Available

Remote Control
Standard Remote Control for classroom use

10/10

+ Borrow This Item

Available

Extension Wire
Standard Extension Wire for classroom use

10/10

+ Borrow This Item

Available

Speaker
Standard Speaker for classroom use

10/10

+ Borrow This Item

Browse Resources

Fig. 6 shows the resource browsing page where faculty can view available items and their quantities.

4. Borrow Request System

Faculty can submit borrowing requests through a digital form. The form includes:

- **Resource Selection:** Choose the item to borrow from a dropdown list.
- **Quantity:** Specify the number of units needed.
- **Borrow Duration:** Set the borrowing period (default: 2 hours).
- **Room / Delivery Location:** Specify where the item should be delivered (e.g., building and room number).
- **Notes:** Optional additional instructions.

Figure 7: Borrow Request Form

ICCT CAINTA
Campus MES

ronjay
Faculty

Browse Resources

My History

Notifications

Logout

Borrow: HDMI Cable
Submit a borrowing request

HDMI Cable
Standard HDMI Cable for classroom use

Available: 10/10

Quantity to borrow
1

Room
e.g., Building, Room

How long will you borrow it?
2 hours

Due: --

Submit Borrow Request

Cancel

Borrow Request Form

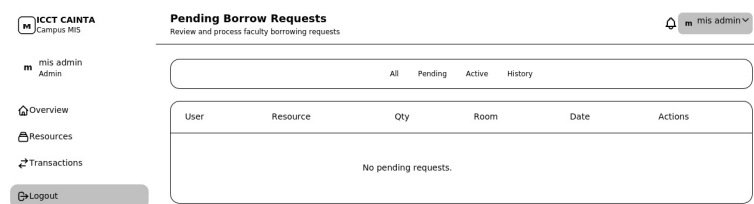
Fig. 7 shows the borrow request form with the Room field where faculty can specify the delivery location.

5. Admin Approval Workflow

When a faculty member submits a borrow request, it appears in the admin's pending requests list. The administrator can:

- **Approve:** Confirms the request and starts the borrowing period.
- **Reject:** Denies the request with an optional reason displayed to the faculty member.

Figure 8: Admin Pending Borrows



Admin Pending Borrows

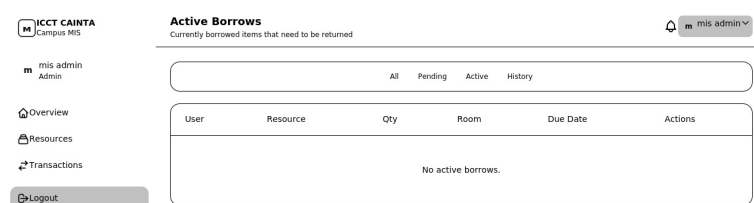
Fig. 8 shows the admin pending borrows list where administrators can approve or reject requests.

6. Active Borrows Tracking

Approved borrows are displayed in the admin's active borrows view, showing:

- Borrower name
- Resource borrowed
- Quantity
- Room / delivery location
- Borrow date and due date/time
- Overdue status indicator

Figure 9: Admin Active Borrows



Admin Active Borrows

Fig. 9 shows the active borrows view with room/location information and due date tracking.

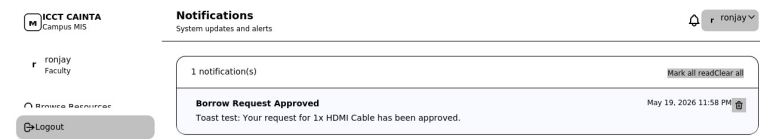
7. Notification System

The system includes a built-in notification system with:

- **Bell Icon:** Displays in the navigation bar with an unread count badge.
- **Real-Time Polling:** Checks for new notifications every 15 seconds.
- **Notification Types:** Approval confirmation, rejection notice, due date reminders (30 minutes before due), and overdue alerts.

- **Notification Page:** Full list of all notifications with read/unread status.

Figure 10: Faculty Notifications



Faculty Notifications

Fig. 10 shows the faculty notifications page with approval and status notifications.

8. Approved Items Banner

When a faculty member's borrow request is approved, a prominent banner appears at the top of their dashboard with the message:

"Your request for [quantity]x [resource] has been approved — your item is on the way."

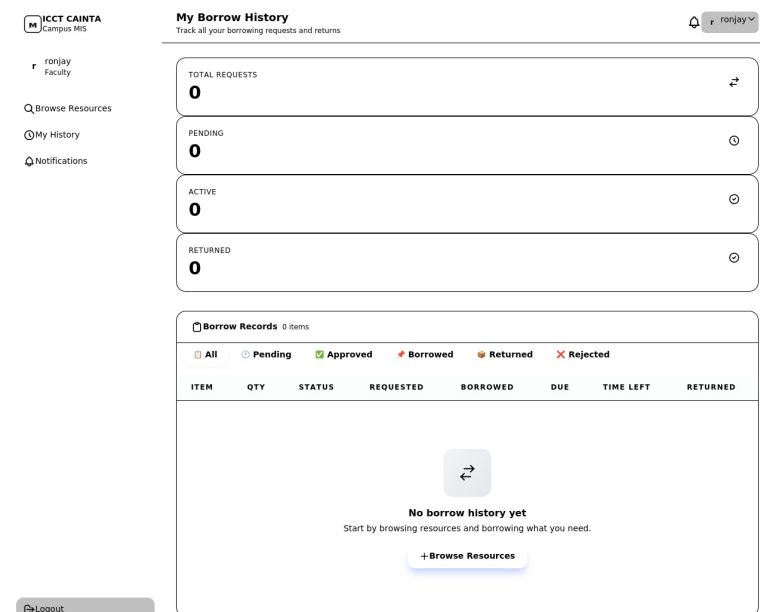
The banner auto-dismisses after 30 seconds and can be manually dismissed. It uses localStorage to prevent reappearing after dismissal until a new approval arrives.

9. Borrow History

Both faculty and administrators can view a searchable history of all borrow transactions. Each record shows:

- Resource name and description
- Quantity
- Room / delivery location
- Date requested
- Status (pending, approved, rejected, returned)
- Date returned (if applicable)

Figure 11: Faculty Borrow History



Faculty Borrow History

Fig. 11 shows the faculty borrow history with status tracking and room/location information.

10. Admin Resource Management

Administrators can manage the resource inventory, including adding new resources, updating quantities, editing resource details, and removing items that are no longer available.

Figure 12: Admin Resource Management

Name	Total Qty	Borrowed	Available	Status	Actions
HDMI Cable	10	0	10	Available	Edit Delete
Remote Control	10	0	10	Available	Edit Delete
Extension Wire	10	0	10	Available	Edit Delete
Speaker	10	0	10	Available	Edit Delete

Admin Resources

Fig. 12 shows the admin resource management page for managing inventory.

C. QR Code Implementation

QR codes are placed on physical resources or at strategic locations such as the resource room entrance. When scanned using any standard QR code scanner (including smartphone camera apps), the QR code redirects the user to the system's login page or directly to the faculty dashboard. This approach provides the following advantages:

- **No special scanner required:** Uses standard smartphone cameras.
- **Simple implementation:** QR codes are generated externally and linked to the system's URL.
- **Flexible placement:** QR codes can be printed and placed on resource shelves, equipment, or room entrances.

The QR code serves as a convenient access shortcut, while the actual authentication and transaction handling are managed through the secure web-based system.

D. Database Design

The system uses a relational database with the following core tables:

- **users:** Stores faculty and administrator account information (name, email, password, role).
- **resources:** Stores resource inventory data (name, description, category, total quantity, available quantity).
- **borrows:** Stores borrowing transaction records (user_id, resource_id, quantity, room, status, borrowed_at, due_at, returned_at, notes).
- **notifications:** Stores system notifications (user_id, title, message, type, is_read, created_at).

V. Conclusion and Recommendations

A. Conclusion

The QR Code-Enabled Web-Based Resource Borrowing and Returning System for MIS Faculty at ICCT Cainta Campus was successfully designed and developed to address the inefficiencies of the traditional manual borrowing and returning process. The system replaces paper-based request letters and physical ID presentation with a streamlined digital workflow that includes QR code access, real-time resource availability tracking, digital borrow request submission with delivery location input, administrator approval workflow, automated notifications, and borrow history tracking.

The system successfully achieves its objectives by:

1. Providing QR code integration that serves as a convenient access point to the web-based system.
2. Displaying real-time resource availability with exact quantities.
3. Enabling digital borrowing request submission with a room or delivery location field.
4. Equipping administrators with tools to approve, reject, and monitor borrowing transactions.
5. Delivering real-time notifications for approval status, due dates, and overdue items directly within the dashboard.
6. Maintaining a searchable borrow history for both faculty and administrators.

By digitizing and automating the borrowing and returning workflow, the system reduces manual workload, improves transaction speed, enhances transparency, and provides better accountability in resource management. The addition of the room or delivery location field addresses the practical need of ensuring borrowed items reach the correct faculty member.

B. Recommendations

Based on the development and implementation of the system, the following recommendations are proposed for future enhancements:

1. **Email Notifications:** Integrate email notifications (e.g., via SendGrid or SMTP) to send alerts outside the system, ensuring faculty are notified even when not logged into the dashboard.
2. **QR Code Generation Module:** Implement a built-in QR code generator within the system that automatically generates and prints QR codes for each resource.
3. **Automated Reports:** Develop a reporting module that generates and exports reports on borrowing frequency, overdue rates, and resource utilization for management review.
4. **Mark as Delivered Feature:** Add a delivery confirmation step so administrators can mark items as “delivered” separately from “returned,” providing more granular tracking.
5. **Overdue Penalties:** Implement a penalty or logging system for overdue returns to encourage timely item returns.
6. **Mobile Application:** Develop a mobile application version for easier on-the-go access and QR scanning.
7. **Multi-Department Scalability:** Extend the system to serve other departments within ICCT, with configurable department-specific

settings.

References

- [1] A. Roy, "Automation Attendance Systems Approaches a Practical Review," Easy Chair Preprint, 2022.
- [2] F. Masalha and N. Hirzallah, "A Students Attendance System Using QR Code," 2014.
- [3] M. A. Rahman, A. K. M. A. Hossain, and M. S. Islam, "Application of QR Code Technology in Higher Education Resource Management," Adv. Inf. Technol. Comput. Sci. (AITCS), 2022.
- [4] K. J. Liew and T. H. Tan, "QR Code-Based Student Attendance System," 2021.
- [5] A. Sharma, P. Kumar, and S. Verma, "Quality Evaluation of Academic Information Systems Using ISO/IEC 25010," Int. J. Innov. Sci. Res. Technol., vol. 9, no. 10, 2024.
- [6] J. Dela Cruz, M. Santos, and R. Reyes, "The Features of Quick Response (QR) Code as an Attendance Monitoring System: Its Acceptability and Implication to Classroom," 2023.
- [7] M. Bautista , "Development of Employee Attendance and Management System Using Quick Response (QR) Code in Sorsogon State University Castilla Campus, Philippines," ResearchGate, 2022.
- [8] C. Ramos , "Study on the Impact and Effectiveness of QR Code and SMS-Based Attendance Monitoring System," 2023.
- [9] R. Mendoza , "Student ID Validation Monitoring System," ResearchGate, 2024.
- [10] L. Gomez , "Assessing the Impact of QR Code Scanning on Attendance Monitoring and Punctuality among Grade 12 Learners," 2024.
- [11] R. Singh, A. Kumar, and S. Sharma, "QR Code-Based Access Control and Resource Management System for Technical Universities," in , 2022, pp. 1-6.
- [12] N. Patel and M. Patel, "Advanced QR Code Based Identity Card: A Novel Approach for Educational Institutions," 2020.
- [13] P. Patel, "Revolutionizing Academic Management: A Digital Approach to Location-Based QR Attendance System and Mobile Application Integration for Lecture and Laboratory Oversight: Smart QR," in , Nov. 2025.
- [14] D. Hariyanto, E. Susanto, and R. Wijaya, "QR Code-Based Student Management System for Login Authentication and Asset Checkout," , vol. 6, no. 1, pp. 45-58, Jun. 2022.
- [15] F. Mahmood, M. Ali, and S. Khan, "Cloud-Integrated QR Code System for Student Login and Resource Management," , vol. 8, no. 1, pp. 89-102, Mar. 2023.
- [16] K. Lopez , "A Web-Based Library Borrowing and Returning System using QR Code Integration and Email Notification for USTP Balubal Campus," ResearchGate, 2025.